

RIYA MITTAL

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EDUCATION

University of Florida

Aug 2024 – May 2026

M.S. in Computer Science · GPA: 3.7 / 4.0

Galgotias University

Dec 2020 – May 2024

B.Tech. in Computer Science · GPA: 3.7 / 4.0

EXPERIENCE

Software Engineer — AI & Automation

May 2025 – Apr 2026

Augmantis.AI (AI division of eComStreet) · Remote · Internship → Part-time

Warranty Information Agent (Azure OpenAI · RAG · NLP)

May – Aug 2025

- Built a **RAG-based warranty retrieval agent** using **Azure OpenAI**, Cosmos DB, and Azure Cognitive Search with vector embeddings, enabling natural-language querying over contracts and claim histories; reduced manual lookup time by 60%.
- Developed **Python (Flask)** backend automation services with modular architecture, logging, and error handling for scalable document processing pipelines.

AI Claims Agent (Anthropic Claude · AWS Bedrock · Svelte)

Sep 2025 – Apr 2026

- Engineered an **end-to-end AI claims agent** using **Anthropic Claude via AWS Bedrock**, automating claim intake, classification, and resolution; projected 10% reduction in average claim processing time.
- Built agentic workflow orchestration — tool-calling, context management, multi-step decision routing — with fallback and escalation logic; developed **Svelte** dashboard for real-time adjuster review and override.
- Implemented **prompt engineering** and retrieval optimization to extend claim categories and improve model accuracy; monitored production reliability via Azure Monitor.

PROJECTS

Non-linear Narrative Therapy System

Jan 2025 – Present

- Designed and built a **multi-path narrative therapy prototype** on Twine, restructuring the story-logic engine to support re-authoring, externalization, and dialogical meaning-making flows used in clinical narrative therapy.
- Engineered a **system-prompt-driven conversational AI layer** using **Azure OpenAI** to generate context-aware reflective dialogue, maintaining session state and user context across non-linear story branches.
- Integrated **LLM-powered narrative generation** with a lightweight full-stack system (Twine frontend + JavaScript state manager + REST API), enabling real-time story adaptation based on user inputs and therapeutic goals.

Multilingual Hate Speech Detection: XLM-RoBERTa vs GPT-5-mini github.com/riiyaaa/Hindi-toxicity-detection

- Fine-tuned **XLM-RoBERTa** for 5-class **multi-label hostile speech detection** (offensive, hate, defamation, fake, non-hostile) on Hindi social media; achieved micro-F1 of 75.9% vs GPT-5-mini zero-shot F1 of 79.6% on binary classification.
- Benchmarked fine-tuned **supervised learning** against **GPT-5-mini zero-shot prompting**, demonstrating trade-offs in precision (GPT: 100%), recall (GPT: 66%), and cultural nuance — revealing key LLM limitations on low-resource Hindi text.
- Applied **Integrated Gradients** and attention rollout for token-level explainability; conducted error analysis on sarcasm, Hinglish code-mixing, and culturally implicit hostility to identify model failure modes.

eBook Reader github.com/riiyaaa/E-Reader

- Built a full-stack eBook reader with **ReactJS frontend** and **Go backend**, integrating **AWS S3** for cloud storage and **WebSockets** for real-time cross-device session sync.
- Implemented personalization features — bookmarks, highlights, reading progress — with optimized rendering pipelines supporting **100+ concurrent file uploads** during load testing with zero data loss.

SKILLS

Languages: Python, C++, Go, JavaScript / TypeScript, SQL

Frameworks & Tools: ReactJS, Svelte, Node.js, Flask, Express, PyTorch, Docker, REST APIs, Git

AI / ML: LLM Agents, RAG, Prompt Engineering, Fine-tuning, NLP, XLM-RoBERTa, Vector Embeddings, Transformers

Cloud & APIs: Azure OpenAI · AWS Bedrock · Anthropic Claude · Azure Cosmos DB · Cognitive Search · AWS S3 · WebSockets

Databases: MongoDB, Cosmos DB, PostgreSQL, SQL

AWARDS & CERTIFICATIONS

- National Winner — Smart India Hackathon 2022:** Led 6-member team; selected from 10,000+ national participants.
- Top 10 — 3 Hackathons:** Prototyped full-stack solutions under 24–48hr deadlines. **Open Source:** 4+ merged PRs to JGEC Winter of Code. **Certs:** Microsoft Azure OpenAI · freeCodeCamp Web Design.